SYSADMIN & IT INFRASTRUCTURE — LAB ASSIGNMENT

File Transfer Across Systems

Windows 10 ↔ Ubuntu 22.04 (VirtualBox)

Course	OS Fundamentals & IT Infrastructure 101
Lab Number	Lab 03
Skill Level	Beginner → Intermediate
Estimated Time	90 – 120 minutes
Environment	Windows 10 (Local) + Ubuntu 22.04 (VirtualBox)
Tools Used	PowerShell, OpenSSH, SCP, WinSCP / FileZilla

LEARNING OUTCOMES

- Configure SSH server on Ubuntu VM and connect from Windows 10 PowerShell
- Transfer files in both directions using SCP from the command line
- Use WinSCP/FileZilla as a GUI alternative to SCP for graphical file management
- Set up a Shared Folder via VirtualBox Guest Additions for seamless host-guest access
- Understand and apply correct Linux file permissions after transfer
- Demonstrate real-world skills that form the foundation of IT support and sysadmin roles

Section 1 — Lab Overview & Real-World Context

In any IT infrastructure or system administration role, moving files securely between different operating systems is a core daily task. Whether you are **deploying configuration files to a Linux server**, pulling log files from a remote machine, or syncing data between a Windows workstation and a Linux host, you need to know how to do this confidently — both via the **terminal (command line)** and via **GUI tools**. This lab teaches all of that in a safe VirtualBox environment.

Why This Matters — Real Scenarios

- An IT support technician collects diagnostic logs from a Linux server to a Windows PC for analysis.
- A sysadmin deploys a new config file from their Windows workstation to a production Ubuntu server.
- A DevOps engineer pulls build artifacts from a Linux CI machine to a Windows machine for packaging.
- A network admin backs up device configs stored on a Linux system to a shared Windows folder.

Three Methods You Will Learn

This lab covers **three progressively powerful methods** for file transfer:

- **Method A —** SCP (Secure Copy Protocol) over SSH using PowerShell and Ubuntu Terminal
- **Method B —** WinSCP / FileZilla GUI tool for graphical drag-and-drop transfer
- **Method C —** VirtualBox Shared Folders (Guest Additions) for seamless host-guest integration

Each method builds on the last. By the end, you will understand **when to use which approach** — exactly the judgement a real IT professional exercises.

Section 2 — Pre-Lab Setup & Requirements

Your Lab Environment

Before starting, confirm the following is in place:

Component	Windows 10 Side	Ubuntu Side
OS	Windows 10 (local machine)	Ubuntu 22.04 LTS in VirtualBox
RAM	At least 4 GB total	Allocate 2 GB to VM
Storage	Local drive (any)	20 GB virtual disk
Network Mode	—	VirtualBox: NAT or Bridged

Component	Windows 10 Side	Ubuntu Side
Terminal	PowerShell 5.1+ or Windows Terminal	GNOME Terminal (Ctrl+Alt+T)
Tools	OpenSSH Client (built-in Win10)	openssh-server package

Step 1 — Find the Ubuntu VM IP Address

You need the IP address of your Ubuntu VM for SSH/SCP connections. Boot your Ubuntu VM, open the terminal, and run:

```
$ ip addr show
# Look for: inet 10.0.2.x   (NAT mode) or 192.168.x.x   (Bridged mode)
# Example output:
# 2: enp0s3: <BROADCAST,MULTICAST,UP>
#     inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
```

NAT vs Bridged — Which Should You Use?

If your VM uses NAT (default), Windows cannot reach the VM directly by IP. You must use Port Forwarding (Rule: Host 127.0.0.1:2222 → Guest 10.0.2.15:22). Bridged mode gives the VM its own IP on your network, making SSH and SCP much simpler. Changing to Bridged is recommended for this lab.

Step 2 — Enable Bridged Networking (Recommended)

In VirtualBox with the Ubuntu VM powered off:

1. Click the Ubuntu VM → Settings → Network
2. Set Attached to: Bridged Adapter
3. Select your Windows Wi-Fi or Ethernet adapter
4. Click OK, then Start the VM
5. Run ip addr show in Ubuntu to confirm a 192.168.x.x address

Section 3 — Method A: SSH & SCP via Terminal

What You Will Learn

Install and configure OpenSSH Server on Ubuntu, connect using PowerShell, and transfer files both ways using the SCP command. This is the most important method — it is how real sysadmins and DevOps engineers move files every day.

Part A1 — Install SSH Server on Ubuntu

On your Ubuntu VM terminal, run the following commands:

```
$ sudo apt update
$ sudo apt install openssh-server -y

# Verify SSH service is running:
$ sudo systemctl status ssh
# You should see:  Active: active (running)

# Enable SSH to start automatically on boot:
$ sudo systemctl enable ssh
```

What is SSH?

SSH (Secure Shell) is an encrypted network protocol that lets you remotely control a machine's terminal and transfer files securely. It replaced older insecure protocols like Telnet and FTP. Port 22 is the default SSH port.

Part A2 — Verify OpenSSH Client on Windows 10

Windows 10 (version 1809+) includes an OpenSSH Client. Verify it in PowerShell (run as Administrator):

```
# Check OpenSSH Client is installed:
PS> Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH.Client*'

# If State shows NotPresent, install it:
PS> Add-WindowsCapability -Online -Name OpenSSH.Client~~~~0.0.1.0

# Confirm SSH is accessible:
PS> ssh -V
```

Part A3 — Connect from Windows PowerShell to Ubuntu via SSH

Replace **student** with your Ubuntu username and **192.168.x.x** with your Ubuntu VM's IP address:

```
# Syntax: ssh username@ip_address
PS> ssh student@192.168.1.50

# First connection: accept the fingerprint
# Type:  yes   then press Enter

# Enter your Ubuntu password when prompted
# Success: prompt changes to   student@ubuntu:~$

# Run a command on Ubuntu FROM PowerShell:
PS> ssh student@192.168.1.50 'uname -a && df -h'

# Exit the SSH session:
student@ubuntu:~$ exit
```

Part A4 — Transfer Files Using SCP

SCP (Secure Copy) uses SSH to transfer files. Syntax: `scp source destination`. The `@` and `:` identify remote paths.

Windows → Ubuntu (Upload)

```
# Create a test file on Windows first:
PS> 'Lab Transfer Test - Windows to Ubuntu' | Out-File
C:\LabFiles\test_upload.txt

# SCP syntax: scp local_file username@ip:remote_path
PS> scp C:\LabFiles\test_upload.txt student@192.168.1.50:/home/student/

# Upload an entire folder (use -r for recursive):
PS> scp -r C:\LabFiles\ProjectDocs
student@192.168.1.50:/home/student/projects/

# Verify on Ubuntu:
student@ubuntu:~$ ls -la ~/
student@ubuntu:~$ cat ~/test_upload.txt
```

Ubuntu → Windows (Download)

```
# Create a test file on Ubuntu first:
student@ubuntu:~$ echo 'Lab Transfer Test - Ubuntu to Windows' >
~/test_download.txt
```

```
# SCP syntax from PowerShell: scp username@ip:remote_file local_path
PS> scp student@192.168.1.50:/home/student/test_download.txt C:\LabFiles\

# Download an entire Ubuntu folder to Windows:
PS> scp -r student@192.168.1.50:/home/student/logs C:\LabFiles\ubuntu_logs\

# Verify on Windows:
PS> Get-Content C:\LabFiles\test_download.txt
```

SCP Quick Reference

`scp [options] source destination`

- **-r** Recursive — copy entire folders
- **-P 2222** Custom port (use with NAT port forwarding)
- **-v** Verbose — show connection details for troubleshooting
- **Remote path format:** `username@ip:/full/path/to/file`

Section 4 — Method B: GUI Transfer with WinSCP / FileZilla

When to Use GUI Tools

GUI tools like WinSCP and FileZilla are ideal when you need to browse the remote file system visually, drag and drop multiple files, or when working in environments where scripting is not yet set up. Many helpdesk and junior sysadmin roles rely on these daily.

Part B1 — Install WinSCP on Windows

WinSCP is a free, open-source SFTP/SCP client for Windows. Download it from winscp.net/eng/download.php and run the installer (choose Typical installation).

Part B2 — Connect WinSCP to Ubuntu VM

Step	Action	Command / Detail
1	**Open WinSCP**	`Launch WinSCP from Start Menu`
2	**New Site**	`Click New Site in the Login dialog`
3	**File Protocol**	`Select: SFTP`
4	**Host name**	`Enter: 192.168.1.50   (your Ubuntu VM IP)`
5	**Port number**	`22`
6	**User name**	`student   (your Ubuntu username)`
7	**Password**	`Enter your Ubuntu account password`
8	**Save & Login**	`Click Save, then Login — Accept key fingerprint`
9	**Connected!**	`Left pane = Windows, Right pane = Ubuntu VM`

Part B3 — Transfer Files in WinSCP

Once connected, WinSCP shows two panels side by side — Windows on the left, Ubuntu on the right. You can:

- Drag and drop files from left (Windows) to right (Ubuntu) to upload
- Drag and drop from right (Ubuntu) to left (Windows) to download
- Right-click any file to see Copy, Move, Delete, Edit, Change Permissions
- Use F5 to copy, F6 to move, F8 to delete selected files

Part B4 — Alternative: FileZilla

FileZilla is cross-platform (Windows, Linux, Mac) and supports SFTP. Download from filezilla-project.org. Connect using:

```
Host:      sftp://192.168.1.50
Username:  student
Password:  (your Ubuntu password)
Port:      22

Click Quickconnect
Left panel = local (Windows), Right panel = remote (Ubuntu)
```

Section 5 — Method C: VirtualBox Shared Folders

What Are Shared Folders?

VirtualBox Shared Folders let you mount a Windows folder directly inside your Ubuntu VM. Files saved there appear instantly on both sides — no network needed. This is ideal for rapid development, editing config files, and local testing workflows.

Part C1 — Install VirtualBox Guest Additions on Ubuntu

Guest Additions must be installed in the Ubuntu VM to enable Shared Folders. With the Ubuntu VM running:

Step	Action	Command / Detail
1	**Insert ISO**	`VirtualBox Menu: Devices → Insert Guest Additions CD Image`
2	**Mount in Ubuntu**	`$ sudo mount /dev/cdrom /media/cdrom`
3	**Run installer**	`$ sudo /media/cdrom/VBoxLinuxAdditions.run`
4	**Reboot VM**	`$ sudo reboot`
5	**Add user to group**	`$ sudo usermod -aG vboxsf $USER  then log out/in`

Part C2 — Configure Shared Folder in VirtualBox Settings

Power off the Ubuntu VM, then in VirtualBox:

Step	Action	Command / Detail
1	**Settings**	`Click Ubuntu VM → Settings → Shared Folders`
2	**Add Folder**	`Click the + icon (Add Shared Folder)`
3	**Folder Path**	`Browse to C:\SharedLabFiles  (create this folder on Windows first)`
4	**Folder Name**	`Type: SharedLabFiles`
5	**Options**	`Check: Auto-mount  and  Make Permanent`
6	**Mount Point**	`Enter: /media/sf_SharedLabFiles`
7	**OK**	`Click OK and Start the VM`

Part C3 — Access the Shared Folder in Ubuntu Terminal

```
# List the shared folder contents from Ubuntu:
$ ls /media/sf_SharedLabFiles/
```

```
# Copy a file FROM Windows share to Ubuntu home:
$ cp /media/sf_SharedLabFiles/report.txt ~/documents/

# Copy a file FROM Ubuntu to the Windows share:
$ cp ~/scripts/backup.sh /media/sf_SharedLabFiles/

# On Windows — files appear instantly at:
# C:\SharedLabFiles\backup.sh
```

On Windows PowerShell, verify the shared files:

```
PS> ls C:\SharedLabFiles
PS> Get-Content C:\SharedLabFiles\backup.sh
```

Section 6 — File Permissions After Transfer

Files transferred from Windows to Linux may arrive with incorrect permissions. Understanding and correcting permissions is a **critical sysadmin skill**.

Linux Permission Basics

Symbol	Meaning	Numeric Value	Example Usage
`r`	Read	4	View file contents
`w`	Write	2	Edit or delete a file
`x`	Execute	1	Run a script or program
`rw-`	Read + Write	6	Typical for data files
`rwx`	Read + Write + Execute	7	Executable scripts
`644`	Owner rw, Group r, Others r	644	Standard file permission
`755`	Owner rwx, Group rx, Others rx	755	Standard script permission

Fixing Permissions After Transfer

```
# View permissions of transferred files:
$ ls -la ~/

# Set correct permissions for a config file (read/write for owner only):
$ chmod 600 ~/transferred_config.txt

# Make a transferred script executable:
$ chmod 755 ~/scripts/backup.sh
$ ./backup.sh

# Change ownership if file was transferred as root:
$ sudo chown student:student ~/transferred_file.txt

# Set permissions recursively on a transferred folder:
$ chmod -R 755 ~/ProjectDocs/
```

Section 7 — Troubleshooting Guide

Troubleshooting is a core IT skill

When something does not work, read the error carefully, work systematically, and eliminate possibilities one at a time. The table below covers the most common issues students encounter in this lab.

Problem	Likely Cause	Fix
`ssh: Connection refused`	SSH server not running on Ubuntu	`sudo systemctl start ssh`
`ssh: Connection timed out`	Wrong IP or NAT blocking port 22	`Verify IP with ip addr show; switch to Bridged mode or set up port forwarding`
`Permission denied (publickey)`	Password auth disabled	`Edit /etc/ssh/sshd_config: set PasswordAuthentication yes, then restart SSH`
`scp: No such file or directory`	Wrong path in command	`Use ls to verify paths; check for typos in remote path`
`WinSCP: Host key mismatch`	VM's SSH key changed (new snapshot)	`In WinSCP: delete stored key, reconnect and accept new key`
`Shared folder not visible in Ubuntu`	Guest Additions not installed or user not in vboxsf group	`Reinstall Guest Additions; run sudo usermod -aG vboxsf $USER and reboot`
`Permission denied on transferred file`	File needs execute or write permission	`chmod 755 filename for scripts; chmod 644 for regular files`
`SCP very slow`	Encryption overhead on slow adapter	`Use FileZilla (SFTP) which can be faster; or use rsync for large transfers`

Section 8 — Assessment & Deliverables

Complete each task and submit the following evidence to your instructor. Screenshots are required where noted.

Task Checklist

#	Task Description	Method	Points
1	Install openssh-server on Ubuntu and confirm systemctl status ssh shows active (running). Screenshot required.	**Ubuntu Terminal**	**10**
2	Verify OpenSSH Client installed on Windows using PowerShell Get-WindowsCapability. Screenshot required.	**PowerShell**	**10**
3	SSH from PowerShell into Ubuntu VM. Run uname -a and whoami inside the SSH session. Screenshot required.	**SSH / PowerShell**	**15**
4	Create a file on Windows (report_win.txt) and SCP upload it to Ubuntu home directory. Verify with ls on Ubuntu.	**SCP Upload**	**15**
5	Create a file on Ubuntu (report_ubuntu.txt) and SCP download it to C:\LabFiles\ on Windows. Verify with Get-Content.	**SCP Download**	**15**
6	Connect to Ubuntu VM using WinSCP or FileZilla GUI. Transfer one file each direction. Screenshot of GUI required.	**WinSCP/FileZilla**	**15**
7	Set up a VirtualBox Shared Folder (C:\SharedLabFiles). Copy a file from Ubuntu to the shared folder and verify it appears on Windows.	**Shared Folders**	**10**
8	After transferring a script (any .sh file) to Ubuntu, set correct permissions with chmod 755. Run the script. Screenshot required.	**Permissions**	**10**

Reflection Questions (Submit Written Answers)

- 1. What is the difference between SCP and SFTP? When would you prefer one over the other?
- 2. Why is it important to disable root login in SSH for production servers?
- 3. Describe a real IT scenario where being able to transfer files between Windows and Linux would be essential.
- 4. What does chmod 600 on a private key file mean, and why is it required by SSH?

Section 9 — Glossary of Key Terms

Term	Definition
SSH (Secure Shell)	Encrypted protocol for remote terminal access and secure file transfer. Default port: 22.
SCP (Secure Copy Protocol)	Command-line tool that uses SSH to transfer files between systems securely.
SFTP (SSH File Transfer Protocol)	A more feature-rich file transfer protocol over SSH. Supports resume, permissions, and more.
OpenSSH	The open-source implementation of the SSH protocol. Includes ssh, scp, and sftp commands.
WinSCP	Free graphical SFTP/SCP client for Windows. Shows local and remote files side by side.
FileZilla	Cross-platform graphical FTP/SFTP client. Works on Windows, Linux, and macOS.
chmod	Linux command to change file permissions. chmod 755 makes a file executable by all users.
chown	Linux command to change file ownership. chown user:group filename.
VirtualBox Guest Additions	Software installed in the VM that enables shared folders, better graphics, and clipboard sharing.
Shared Folder	A host folder mounted inside a VM, visible on both sides for seamless file access.
NAT (Network Address Translation)	Default VirtualBox networking. VM shares the host's IP. Requires port forwarding for SSH.
Bridged Adapter	VirtualBox network mode where the VM gets its own IP on the local network. Simplest for SSH.
Port 22	The default TCP port used by SSH. Must be open in firewalls for SSH connections to work.
Firewall (UFW)	Ubuntu's Uncomplicated Firewall. Run sudo ufw allow ssh to permit SSH connections.
rsync	Advanced file sync tool over SSH. Better than SCP for large files or incremental backups.